

CanYouSee

⚠ **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

How about some hide and seek?

Hints

- How can you view the information about the picture?
 - If something isn't in the expected form, maybe it deserves attention?
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Tools Used

- **unzip**: Extract files from ZIP archives
 - **file**: Utility to identify file types
 - **strings**: Command-line tool to extract readable ASCII sequences from files
 - **grep**: Text search utility for filtering output
 - **exiftool**: Metadata inspection tool
 - **base64**: Encoding/decoding utility for Base64 data
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Complete Solution

Step 1: Download and Prepare

Download the file named `unknown.zip` from the challenge page.

Once downloaded, extract the contents:

```
unzip unknown.zip
```

You'll find an image file.



Step 2: Basic File Inspection

First, verify the file type and general information:

```
file ukn_reality.jpg
```

Expected output:

```
ukn_reality.jpg: JPEG image data, JFIF standard 1.01, resolution (DPI), density 72x72, segment length 16, baseline, precision 8, 4308x2875, components 3
```

Step 3: Search for Visible Text

The flag may be present as plain text. Try extracting all character strings:

```
strings ukn_reality.jpg | grep -i "picoCTF"
```

⚠ *If no results appear, proceed to the next step (decoding).*

Step 4: Inspect Metadata

Use `exiftool` to examine the file metadata:

```
exiftool ukn_reality.jpg
```

Complete output:

```
ExifTool Version Number      : 13.50
File Name                    : ukn_reality.jpg
Directory                    : .
File Size                    : 2.3 MB
File Modification Date/Time   : 2024:03:12 02:05:55+02:00
File Access Date/Time        : 2026:05:08 16:02:59+02:00
File Inode Change Date/Time   : 2026:05:08 16:02:43+02:00
File Permissions              : -rw-r--r--
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
JFIF Version                 : 1.01
Resolution Unit              : inches
X Resolution                  : 72
Y Resolution                  : 72
XMP Toolkit                   : Image::ExifTool 11.88
Attribution URL              : cGljb0NURntNRTc0RDQ3QV9ISUREM05fNmE5ZjVhYzR9Cg==
Image Width                  : 4308
Image Height                  : 2875
Encoding Process              : Baseline DCT, Huffman coding
Bits Per Sample               : 8
Color Components              : 3
Y Cb Cr Sub Sampling         : YCbCr4:2:0 (2 2)
Image Size                   : 4308x2875
Megapixels                   : 12.4
```

 **Key point:** Notice the `Attribution URL` field which contains a suspicious, encoded string!

Step 5: Decode the Base64 Value

The string in the `Attribution URL` field is encoded in Base64:

```
cGljb0NURntNRTc0RDQ3QV9ISUREM05fNmE5ZjVhYzR9Cg==
```

Decode it using one of these commands:

```
echo 'cGljb0NURntNRTc0RDQ3QV9ISUREM05fNmE5ZjVhYzR9Cg==' | base64 -d
```

✅ Final Result

picoCTF{<REDACTED>}

🧠 Key Concepts

- **Metadata:** Hidden information in files
- **Base64:** Common encoding to obscure or compress content
- **Anomaly Recognition:** Looking for elements that seem suspicious or out of place